

Pendulum Modeling and Step Response for TFs

EGR 345

GVSU

Outline

- review: transfer functions
- pendulum modeling
 - we need this for lab
- step function and step response

Announcement

- HW 2 is up

Review: What is a transfer function?

- how do we find them?

$\mathcal{L}[\text{diff. eqn}] \xrightarrow{\text{algebra}} G(s)$

- what can we do with them?

$\rightarrow$ find response to any input

for Mechanical system

FBD $\rightarrow$ Newton's 2nd law $\rightarrow$ eqn of motion

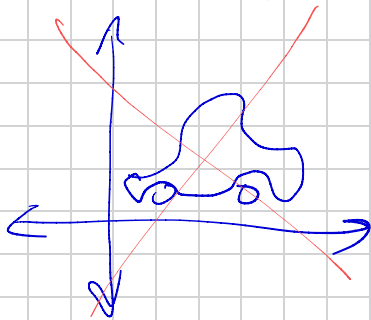
$G(s) \leftarrow$ rearrange
(algebra)

Can you plot a transfer function?

→ No.

$$m\ddot{x} + b\dot{x} + kx = f$$

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$



→ we plot the response of a TF to a specific input

A TF will have different outputs
to different inputs (responses)

$$Out(s) = G(s) \cdot In(s)$$



out(t) depends on in(t)

I use a TF to find the output for any input

$$O_w(s) = G(s) \cdot Y(s)$$

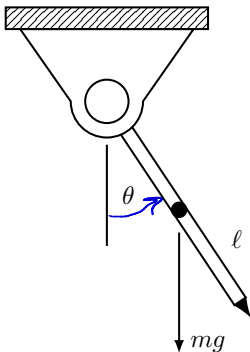
by hand: PFE $\xrightarrow{\mathcal{L}^{-1}}$ out(t)

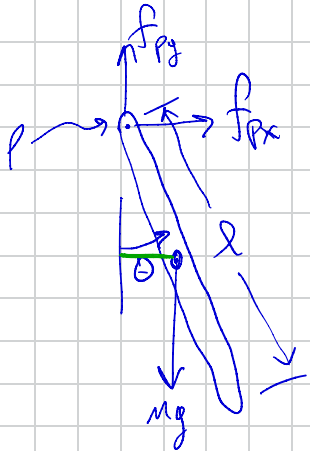
Python:

- control.impulse-response
- control.step-response
- control.forced-response

Pendulum Modeling

- How do we find the equation of motion for this system?





ΣM about P

$$\Sigma M = I\alpha = I\ddot{\theta}$$

$$\Sigma M_P: I\ddot{\theta} = -\frac{l}{2}(\sin\theta)mg$$

$$I\ddot{\theta} + \frac{l}{2}(\sin\theta)mg = 0$$

Taylor Series: $\sin\theta \approx \theta \left[\frac{-\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right]$

ignore for small θ

$$\frac{I}{\uparrow} \ddot{\Theta} + \frac{l}{2} mg \Theta = 0 \rightarrow \frac{1}{3} ml^2 \ddot{\Theta} + \frac{l}{2} mg \Theta = 0$$

$\frac{1}{3} ml^2$ rod rotating about one end

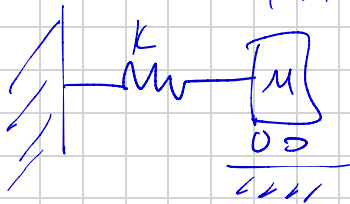
$$\frac{1}{12} ml^2 + m \left(\frac{l}{2} \right)^2 = \frac{1}{3} ml^2$$

$$\left(\frac{1}{3} m l^2 \ddot{\Theta} + \frac{l}{2} m g \Theta = 0 \right) \frac{3}{l^2}$$

$$\ddot{\Theta} + \underbrace{\frac{3g}{2l}}_{\omega_n^2} \Theta = 0$$

$\hookrightarrow x$

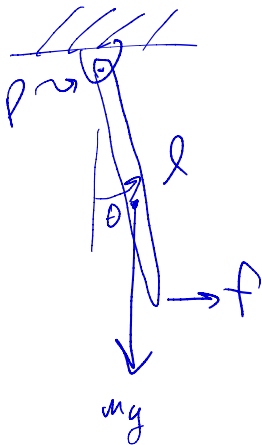
← probably need damping
to account for energy
loss



$$\ddot{x} + \omega_n^2 x = 0$$

Pendulum Modeling 2

- what if there is a force input at the tip of the pendulum?



$$\sum \mathcal{M}_P: I \ddot{\Theta} = -mg \frac{l}{2} \sin \Theta + fl$$

$$I \ddot{\Theta} + mg \frac{l}{2} \sin \Theta = fl$$

$$\left[\frac{1}{3} ml^2 \ddot{\Theta} + \frac{mg l}{2} \Theta = fl \right]$$

$$\left[\frac{1}{3} m l^2 \ddot{\theta} + \frac{m g l}{2} \theta = f l \right] \frac{3}{m l^2}$$

$$\mathcal{L} \left[\ddot{\theta} + \frac{3g}{2l} \theta = f \frac{3}{ml} \right]$$

$$s^2 \theta + \frac{3g}{2l} \theta = F \frac{3}{ml}$$

$$\theta \left(s^2 + \frac{3g}{2l} \right) = f \frac{3}{ml}$$

$$\frac{\theta(s)}{F(s)} = \frac{3/ml}{s^2 + \frac{3g}{2l}}$$

$$= \frac{N}{s^2 + \omega^2}$$

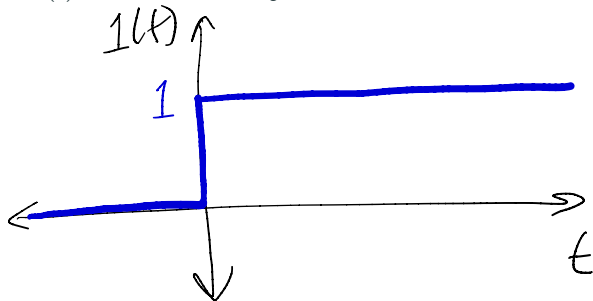
Pendulum TF

$$G(s) = \frac{\Theta(s)}{P(s)} = \frac{3/ul}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

→ curve fit using
control. impulse-response

What is a step function?

- how is $1(t)$ different from just the number 1?



a signal with amplitude 1 that switches "on" at $t=0$

→ when the step input switches on, it causes a transient response

What is a step response?

- and how do we find it?
 - “by hand”
 - with python

↑
control.step-response

$$\mathcal{L}[1(t)] = \frac{1}{s}$$

$$Y_{\text{step}}(s) = G(s) \cdot \frac{1}{s}$$

↑
step response

↑
step input

$$G(s) = \frac{25}{s^2 + s + 25} = \frac{Y(s)}{U(s)}$$

Find the unit step response $\rightarrow y(t)$

$$u(t) = 1(t) \rightarrow U(s) = \frac{1}{s}$$

$$Y(s) = G(s) \cdot U(s) = \underbrace{\left(\frac{25}{s^2 + s + 25} \right)}_{G(s)} \underbrace{\frac{1}{s}}_{U(s)}$$

$$Y(s) = \frac{25}{s^2 + s + 25} \cdot \frac{1}{s} = \frac{C_1}{s} + \frac{C_2 s + C_3 (s+a)}{(s+a)^2 + \omega^2}$$

$$D = s(s^2 + s + 25)$$

roots 8 ± 9

$$s^2 + s + 25 = s^2 + 2as + a^2 + \omega^2$$

$$a = 0.5$$

$$\omega = \sqrt{25 - 0.5^2} = 4.9$$

$$Y(s) = \frac{25}{s^2 + s + 25} \cdot \frac{1}{s} = \frac{C_1}{s} + \frac{C_2 \omega + C_3 (s + a)}{(s + a)^2 + \omega^2} \quad (1)$$

$$25 = C_1 (s^2 + s + 25) + C_2 s \omega + C_3 s (s + a)$$

$$25 = \underline{C_1 s^2} + \underline{C_1 s + 25 C_1} + \underline{C_2 s \omega} + \underline{C_3 s^2 + C_3 s a}$$

$$s^0: 25 = 25 C_1 \rightarrow C_1 = 1$$

$$s^1: 0 = C_1 + C_2 \omega + C_3 a \rightarrow 0 = 1 + C_2 \omega + C_3 a$$

$$s^2: 0 = C_1 + C_3 = C_3 = -C_1 = -1$$

$$C_2 = \frac{a-1}{\omega}$$

Always Check Your Assumptions

$$G(s) = \frac{1}{s^2 + 75s + 25}$$

$$Y(s) = \frac{1}{s} \cdot \frac{1}{s^2 + 75s + 25} \stackrel{?}{=} \frac{C_1}{s} + \frac{C_2 s + C_3(s+a)}{(s+a)^2 + \omega^2}$$

over damped
→ 2 distinct real roots

$$= \frac{C_1}{s} + \frac{C_2}{s+a_1} + \frac{C_3}{s+a_2}$$

$$s^2 + 75s + 25 = (s+a)^2 + \omega^2$$

$$s^2 + 75s + 25 = s^2 + \underline{2as} + \underline{a^2} + \underline{\omega^2}$$

$$a = \frac{75}{2} = 37.5$$

$$25 = a^2 + \omega^2 \rightarrow \omega^2 = 25 - 37.5^2$$

ω is imaginary
 $\rightarrow$ something went wrong